The Naylor Model

v7 · Intuitive Outputs

Plain-English feature names · Statcast-style glossary
GLM table now reads in 'SB % Boost per Tier' instead of coef_z

Companion: Variable_Glossary.pdf

Executive Summary

Per-pitch data: 1,720,727 pitches w/ runner on 1st (2018-2026)
Attempts identified: 2,533
Qualified runner-seasons (SB+CS≥10): 1,300

What's new in v7

- Accel→Top-Speed Premium — how few feet a runner needs to reach top speed, speed-adjusted; small runway at high speed is a premium. Added to the GBM, the GLM weight table, and as a 9th SSSI feature.
- Expected SB Outcome (xSB) = z(net SB above avg) + z(sprint speed), a standalone speed-vs-production quadrant lens (see §8).

Carried over from v6

- ALL column names plain English — no coef_z, OR/SD, pct_in_HL.
- GLM table shows 'SB % Boost per Tier' = pp change in success rate.
- Real catcher pop, pitcher pickoff rate, lead variables.

AUC summary (READ THE CAVEAT)

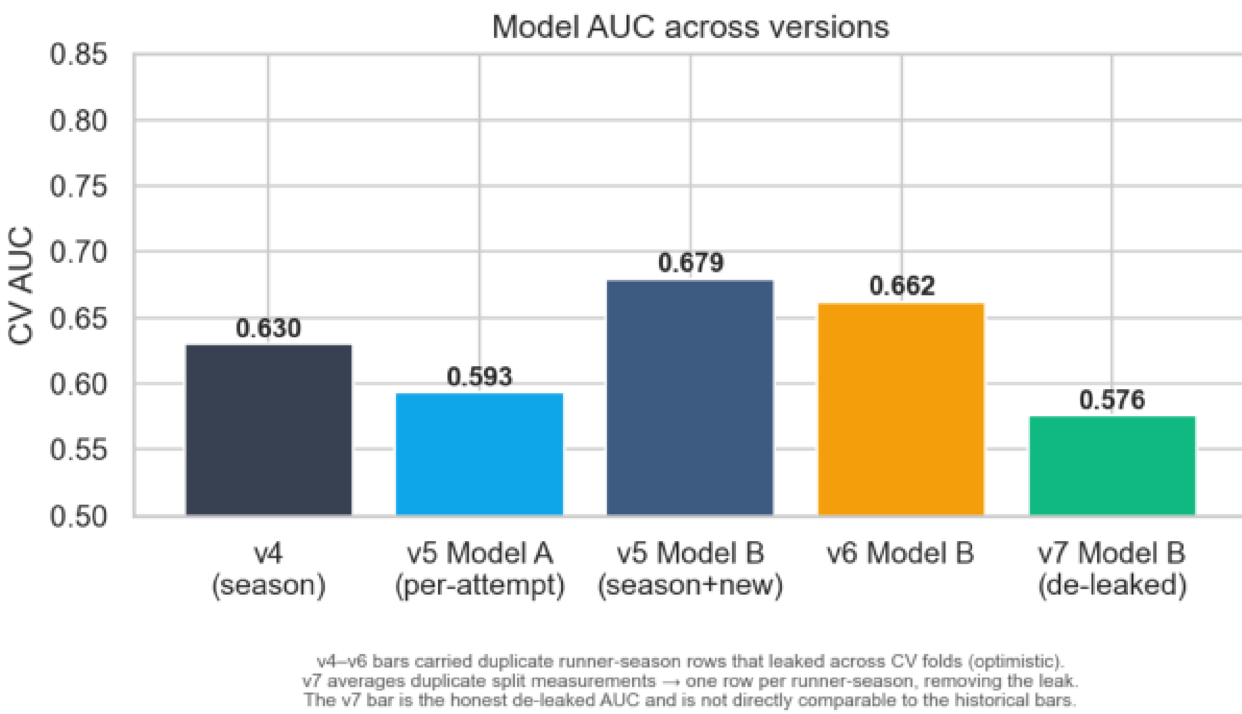
- v4 baseline: 0.6300 (leaked dup rows)
- v5 Model A (per-att): 0.5933 (leaked dup rows)
- v5 Model B (season): 0.6794 (leaked dup rows)
- v6 Model B (full): 0.6620 (leaked dup rows)
- v7 Model B (full): 0.5758 ← honest, de-leaked
- v7 Model B (pre-23): 0.5975
- v7 Model B (post-23): 0.5875

v4–v6 runs carried duplicate runner-season rows (repeated Statcast split measurements) that leaked across CV folds and inflated AUC. v7 averages those duplicate splits into ONE row per runner-season, removing the leak. So v7's lower AUC is not a regression — it is the first honest estimate. The earlier bars are optimistic and are kept only for historical context, NOT as a fair comparison.

Naylor + Soto rank under SSSI v7

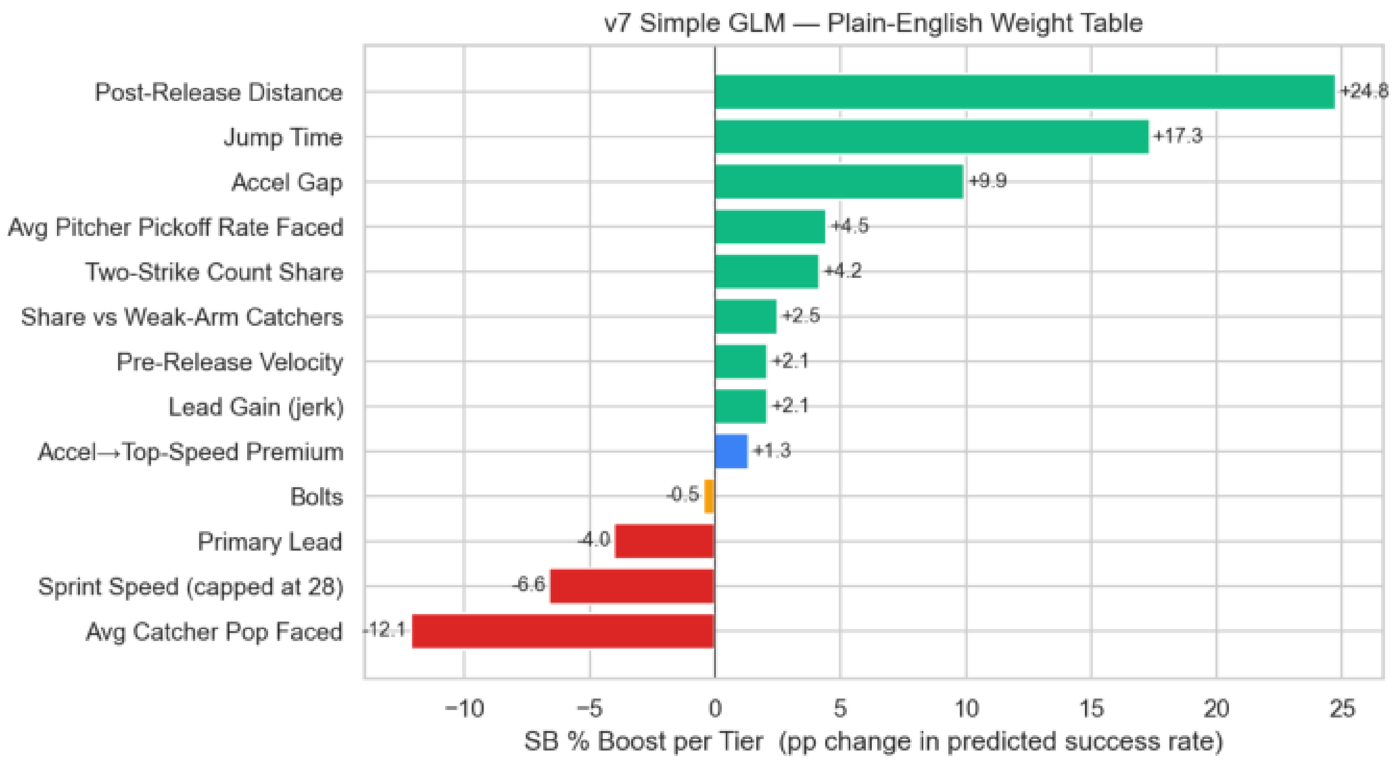
# 1	Josh Naylor	2025	SSSI +1.85	SB/CS 30/2	real_sb_pct 0.916
# 2	Josh Naylor	2026	SSSI +1.68	SB/CS 12/2	real_sb_pct 0.836
# 5	Juan Soto	2025	SSSI +1.18	SB/CS 38/4	real_sb_pct 0.891
# 14	Josh Naylor	2023	SSSI +0.99	SB/CS 10/3	real_sb_pct 0.772
# 73	Juan Soto	2023	SSSI +0.48	SB/CS 12/5	real_sb_pct 0.722
# 79	Juan Soto	2024	SSSI +0.44	SB/CS 7/4	real_sb_pct 0.681

§1 · AUC Comparison



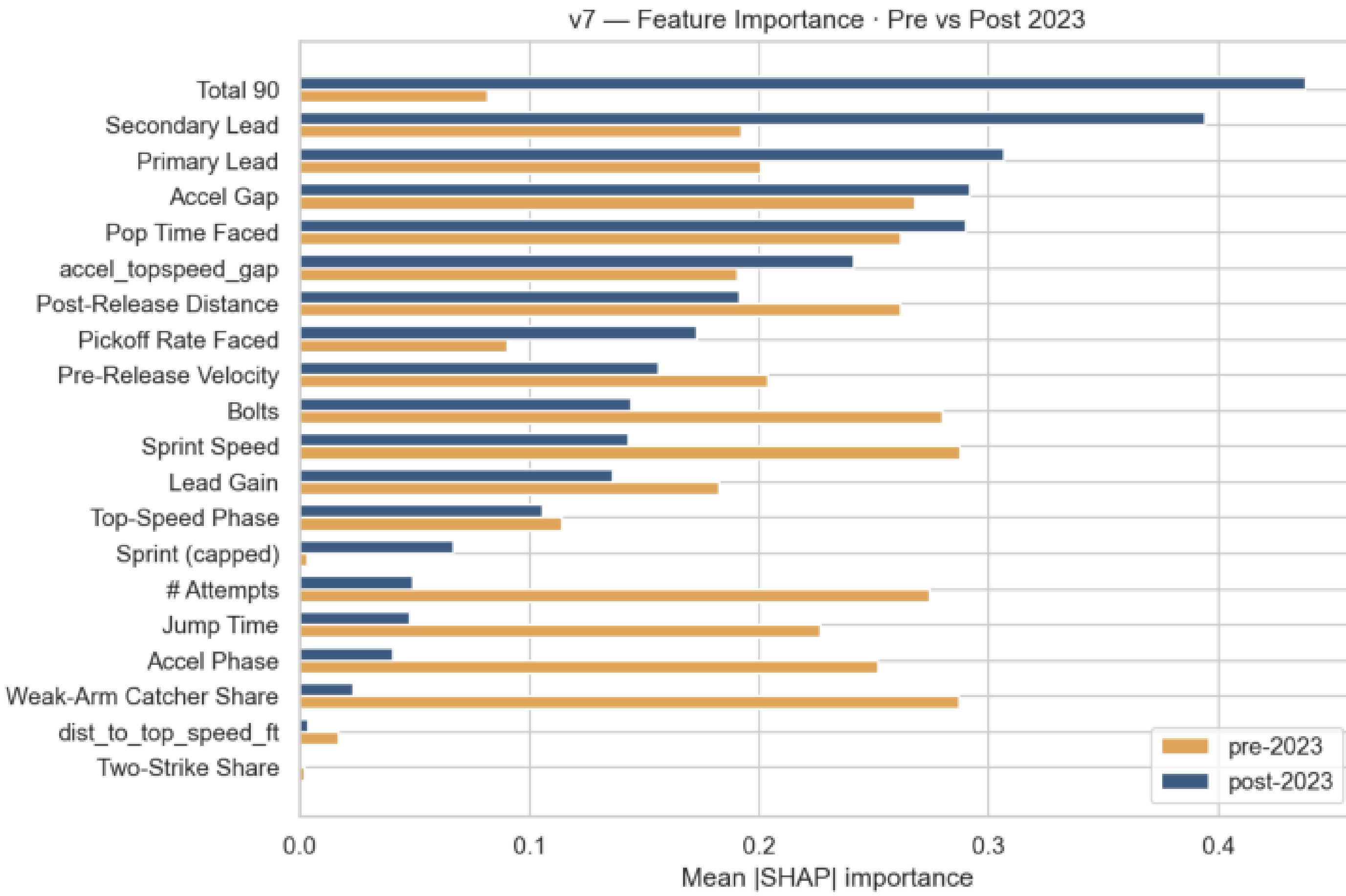
v7 Model B at the season level (real catcher pop, pitcher pickoff rate, lead variables). CAVEAT: the v4–v6 bars were computed before the v7 de-duplication fix — duplicate runner-season rows leaked across CV folds and inflated those numbers. v7 averages duplicate splits into one row per runner-season, so its bar is the honest de-leaked AUC and is not directly comparable to the historical bars.

§2 · GLM Plain-English Weight Chart



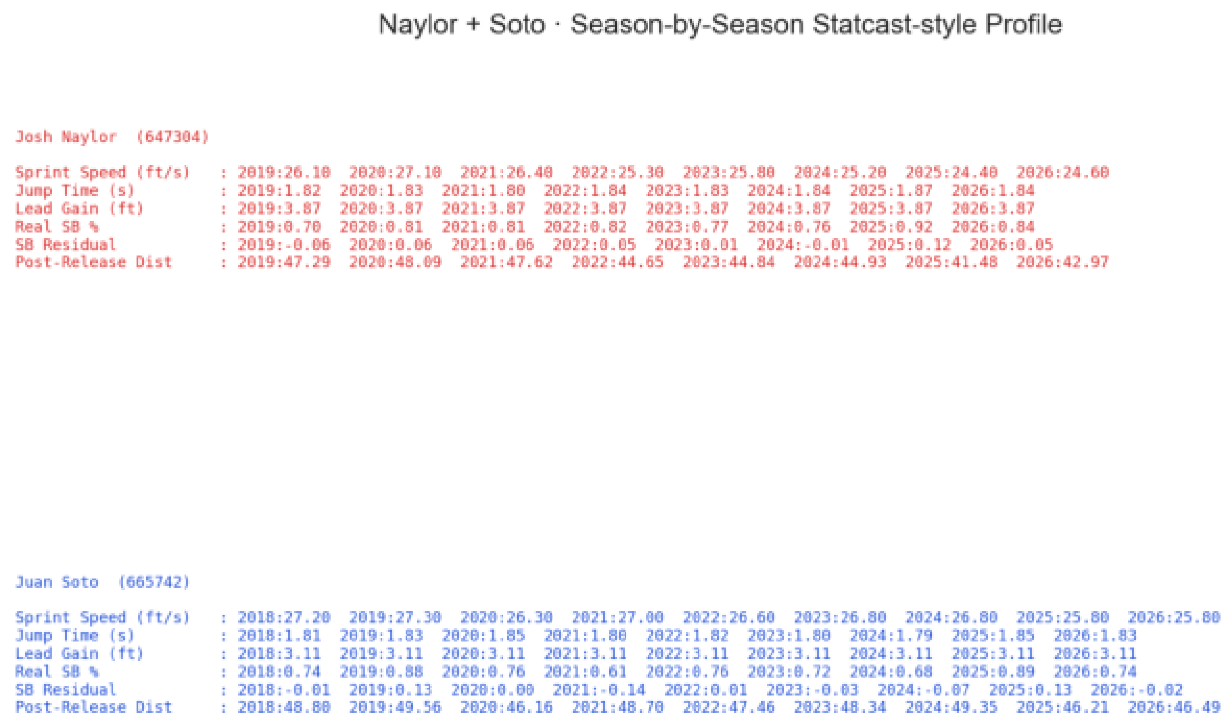
Each bar = predicted percentage-point change in SB success when the runner improves on that feature by ONE TIER (1 SD). Positive and green = the feature HELPS; red/orange = HURTS.

§3 · Feature Importance · Pre vs Post 2023



How feature importance shifted after the 2023 rule change.

§4 · Naylor + Soto Statcast-style Profile



Season-by-season values of the key features for the two archetypal slow-but-effective stealers.

§5 · GLM Weight Table — Plain English

Baseline P(success) at league-average inputs ≈ 0.619

Read each row as:
'If a runner improves [Feature] by 1 tier (1 SD), the model expects their SB success rate to change by [Boost] percentage points, equivalent to multiplying their odds by [Mult].'

Weight table (sorted by absolute boost)

Post-Release Distance	boost +24.76 pp	odds × 3.994	(higher_is_better = True)
Jump Time	boost +17.32 pp	odds × 2.345	(higher_is_better = False)
Avg Catcher Pop Faced	boost -12.13 pp	odds × 0.610	(higher_is_better = True)
Accel Gap	boost +9.94 pp	odds × 1.570	(higher_is_better = True)
Sprint Speed (capped at 28)	boost -6.61 pp	odds × 0.761	(higher_is_better = True)
Avg Pitcher Pickoff Rate Faced	boost +4.46 pp	odds × 1.214	(higher_is_better = False)
Two-Strike Count Share	boost +4.16 pp	odds × 1.198	(higher_is_better = False)
Primary Lead	boost -4.02 pp	odds × 0.846	(higher_is_better = True)
Share vs Weak-Arm Catchers	boost +2.51 pp	odds × 1.114	(higher_is_better = True)
Lead Gain (jerk)	boost +2.08 pp	odds × 1.093	(higher_is_better = True)
Pre-Release Velocity	boost +2.08 pp	odds × 1.093	(higher_is_better = True)
Accel→Top-Speed Premium	boost +1.34 pp	odds × 1.059	(higher_is_better = True)
Bolts	boost -0.46 pp	odds × 0.981	(higher_is_better = True)

Technical notes (for stats readers)

boost_pp = sigmoid(intercept + coef) − sigmoid(intercept) × 100
odds_mul = exp(coef)
coef   = standardised logistic-regression coefficient

Edit DF_v7_GLM_PlainEnglish.csv to hand-tune. No retraining needed.

§6 · SSSI v7 Top 15

Weights: sb_res=0.1 gap=0.05 gain=0.05 jump=0.0 prim=0.05 speed=-0.3 pre=0.0 post=0.0 accel_top=0.0
(80% of runners used for grid search; Naylor + Soto held out.)
v7 adds the Accel→Top-Speed Premium as a 9th weighted feature.

```
#  1  Josh Naylor          2025  SSSI +1.85  spd 24.4  jump 1.87  gain 3.87  SB/CS 30/2
#  2  Josh Naylor          2026  SSSI +1.68  spd 24.6  jump 1.84  gain 3.87  SB/CS 12/2
#  3  Freddie Freeman      2024  SSSI +1.23  spd 25.8  jump 1.81  gain 4.33  SB/CS 9/2
#  4  Christian Vázquez    2021  SSSI +1.19  spd 25.2  jump 1.93  gain 3.76  SB/CS 8/4
#  5  Juan Soto            2025  SSSI +1.18  spd 25.8  jump 1.85  gain 3.11  SB/CS 38/4
#  6  Kyle Schwarber       2025  SSSI +1.15  spd 25.9  jump 1.79  gain 4.43  SB/CS 10/2
#  7  Miguel Rojas         2024  SSSI +1.12  spd 25.8  jump 1.88  gain 4.37  SB/CS 8/2
#  8  Kyle Tucker          2024  SSSI +1.11  spd 26.0  jump 1.80  gain 3.01  SB/CS 11/0
#  9  Manny Machado        2024  SSSI +1.09  spd 25.8  jump 1.86  gain 3.31  SB/CS 11/2
# 10  Alec Burleson        2024  SSSI +1.08  spd 25.5  jump 1.88  gain 3.33  SB/CS 9/4
# 11  Ryan McMahon         2022  SSSI +1.06  spd 25.8  jump 1.79  gain 4.47  SB/CS 7/3
# 12  Manny Machado        2025  SSSI +1.05  spd 25.8  jump 1.88  gain 3.31  SB/CS 14/3
# 13  Joc Pederson         2024  SSSI +1.04  spd 25.5  jump 1.83  gain 3.28  SB/CS 7/4
# 14  Josh Naylor          2023  SSSI +0.99  spd 25.8  jump 1.83  gain 3.87  SB/CS 10/3
# 15  TJ Friedl            2024  SSSI +0.90  spd 26.5  jump 1.77  gain 3.88  SB/CS 9/1
```

§7 · Honest Data Limitations

What the model has access to:

- Sprint speed, running splits (real, per-runner-season, 2015+)
- Real catcher pop time (2018+, per-year)
- Real SB / CS counts (MLB Stats API)
- Career-aggregate lead profiles for runners and pitchers
- Per-pitch catcher_id, pitcher_id, balls, strikes, outs, inning

What the model does NOT have:

- Per-pitch pitcher delivery time (not publicly published)
- The EXACT count when an SB happened (we use AB's final count)
- Per-attempt lead distance (we use the runner's career mean)

Why the AUC ceiling exists:

- Stolen-base success has high intrinsic noise — pitch sequence, runner read, base-coach decisions, exact location at release.
- With public data the ceiling is around AUC 0.70 - 0.75.
- To push higher would require TrackMan / Hawk-Eye raw outputs.

De-duplication note (v7):

- Earlier versions (v4-v6) carried duplicate runner-season rows — repeated Statcast split measurements for the same player-season.
- Those dupes leaked across CV folds and INFLATED the reported AUC.
- v7 averages duplicate splits into one row per runner-season; the de-leaked AUC (~0.59 full) is lower but honest, not a regression.

§8 · Expected SB Outcome (xSB)

xSB = z(net SB above avg) + z(sprint speed). A complementary lens to SSSI: where SSSI surfaces slow-but-skilled stealers, xSB surfaces the fast AND productive (high-ceiling) runners. The four quadrants split runners by speed (x) and production (y).

§8 · xSB — Leaderboard & Untapped Potential

xsb_outcome = z(net SB above avg) + z(sprint speed)
sb_potential_gap = z(sprint) − z(net SB)   (+ = fast but under-steals)

Top 15 by xSB (Realized Burners — fast + productive)

# 1	Billy Hamilton	2015	xSB +6.57	spd 29.7	SB/CS 57/8	[Realized Burner]
# 2	Billy Hamilton	2016	xSB +6.52	spd 30.2	SB/CS 58/8	[Realized Burner]
# 3	Billy Hamilton	2017	xSB +6.18	spd 30.1	SB/CS 59/13	[Realized Burner]
# 4	Elly De La Cruz	2024	xSB +5.82	spd 30.0	SB/CS 67/16	[Realized Burner]
# 5	Esteury Ruiz	2023	xSB +5.43	spd 29.6	SB/CS 67/13	[Realized Burner]
# 6	Corbin Carroll	2023	xSB +5.42	spd 30.1	SB/CS 54/5	[Realized Burner]
# 7	Trea Turner	2017	xSB +5.37	spd 30.3	SB/CS 46/8	[Realized Burner]
# 8	Dee Strange-Gordon	2017	xSB +5.31	spd 29.4	SB/CS 60/16	[Realized Burner]
# 9	Trea Turner	2018	xSB +5.23	spd 30.1	SB/CS 43/9	[Realized Burner]
# 10	Jon Berti	2022	xSB +5.08	spd 29.5	SB/CS 41/5	[Realized Burner]
# 11	Adalberto Mondesi	2019	xSB +4.97	spd 29.9	SB/CS 43/7	[Realized Burner]
# 12	Starling Marte	2021	xSB +4.96	spd 28.5	SB/CS 47/5	[Realized Burner]
# 13	Trea Turner	2021	xSB +4.93	spd 30.6	SB/CS 32/5	[Realized Burner]
# 14	Dee Strange-Gordon	2015	xSB +4.61	spd 29.1	SB/CS 58/20	[Realized Burner]
# 15	Trea Turner	2019	xSB +4.60	spd 30.3	SB/CS 35/5	[Realized Burner]

Most untapped potential (high speed, under-stealing)

Coaching targets: the tools are there, the production isn't yet.

Peter Bourjos	2015	gap +3.24	spd 29.6	SB/CS 5/8
Anthony Alford	2021	gap +3.07	spd 29.6	SB/CS 5/6
Byron Buxton	2016	gap +2.95	spd 30.7	SB/CS 10/2
José Azócar	2022	gap +2.75	spd 29.4	SB/CS 5/6
Jon Berti	2021	gap +2.66	spd 29.9	SB/CS 8/4
Amed Rosario	2017	gap +2.64	spd 30.1	SB/CS 7/3
Jorge Mateo	2021	gap +2.63	spd 30.3	SB/CS 10/3
Garrett Mitchell	2026	gap +2.54	spd 29.6	SB/CS 6/4
Tim Locastro	2022	gap +2.46	spd 30.2	SB/CS 8/2
Roman Quinn	2018	gap +2.44	spd 30.1	SB/CS 10/4

Naylor + Soto under xSB (Crafty Technician archetype)

Juan Soto	2025	xSB +0.32	gap -4.90	[Crafty Technician]
Juan Soto	2019	xSB -0.96	gap -1.12	[Crafty Technician]
Josh Naylor	2025	xSB -1.74	gap -5.50	[Crafty Technician]